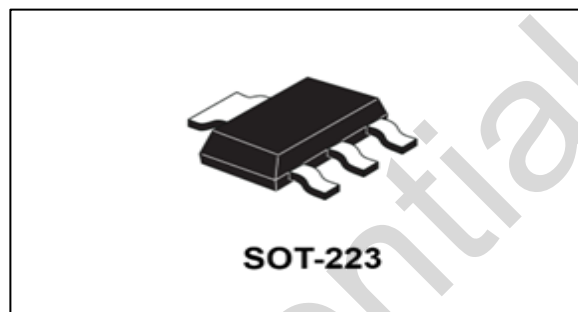


Features

- Linear current limitation
- Thermal shutdown
- Short circuit protection
- Integrated clamp
- Low current drawn from input pin
- Diagnostic feedback through input pin
- ESD protection
- Direct access to the gate of the Power MOSFET (analog driving)
- Compatible with standard Power MOSFET
- Automotive qualified(AEC-Q100 Grade 1)



Description

The HELS120 is intended for replacement of standard Power MOSFETs in up to 50 kHz DC applications.

Built-in thermal shutdown, linear current limitation and overvoltage clamp protect the chip in harsh environments.

Fault feedback can be detected by monitoring voltage at the input pin.

Type	R _{DS(on)}	I _{lim}	V _{clamp}
HELS120-SOT223	120mΩ	2.3A	40V

Part number	Package	Marking	Packing
HELS120RCQC	SOT-223	HELS120	2500 / Tape & Reel

**Not sure if you got the latest datasheet or not. Please kindly contact the sales of KKChips to confirm. Thanks.*

Contents

Features	1
Description	1
1 Block diagram and pin description	5
1.1 Block diagram	5
2 Electrical specifications	6
2.1 Absolute maximum ratings	6
2.2 Thermal data	6
2.3 Electrical characteristics	7
2.4 Electrical characteristics curves	12
3 Protection features	16
4 Package and packing information	17
4.1 SOT-223 mechanical data	17
5 Recommended furnace temperature curve	19
6 Recommended pad size	20
7 Customer Service	21
Revision history	22

List of tables

Table 2-1 Absolute maximum ratings.....	6
Table 2-2 Thermal data.....	6
Table 2-3 Off	7
Table 2-4 On	7
Table 2-5 Dynamic ($T_j = 25^{\circ}\text{C}$, unless otherwise specified)	7
Table 2-6 Switching ($T_j = 25^{\circ}\text{C}$, unless otherwise specified)	7
Table 2-7 Source drain diode ($T_j = 25^{\circ}\text{C}$, unless otherwise specified).....	9
Table 2-8 Protections ($-40^{\circ}\text{C} < T_j < 150^{\circ}\text{C}$, unless otherwise specified).....	9
Table 4-1 SOT-223 mechanical data	18

List of figures

Figure 1-1 Block diagram.....	5
Figure 1-2 Configuration diagram (top view).....	5
Figure 2-1 Current and voltage conventions.....	6
Figure 2-2 Switching time test circuit for resistive load	10
Figure 2-3 Test circuit for diode recovery times.....	10
Figure 2-4 Unclamped inductive load test circuits	11
Figure 2-5 Input charge test circuit	11
Figure 2-6 Unclamped inductive waveforms	11
Figure 2-7 Thermal impedance for SOT-223.....	12
Figure 2-8 Derating curve	12
Figure 2-9 Transconductance	12
Figure 2-10 Static drain-source on resistance vs input voltage (part 1/2).....	12
Figure 2-11 Static drain-source on resistance vs input voltage (part 2/2).....	13
Figure 2-12 Source-drain diode forward characteris.....	13
Figure 2-13 Static drain source on resistance	13
Figure 2-14 Turn-on current slope (part 1/2)	13
Figure 2-15 Turn-on current slope (part 2/2)	13
Figure 2-16 Transfer characteristics	13
Figure 2-17 Static drain-source on resistance vs I_d	14
Figure 2-18 Input voltage vs input charge.....	14
Figure 2-19 Turn-off drain source voltage slope (part 1/2)	14
Figure 2-20 Turn-off drain source voltage slope (part 2/2)	14
Figure 2-21 Capacitance variations.....	14
Figure 2-22 Output characteristics temperature.....	14
Figure 2-23 Normalized on resistance vs temperature	15
Figure 2-24 Switching time resistive load (part 1/2)	15
Figure 2-25 Switching time resistive load (part2/2)	15
Figure 2-26 Normalized input threshold voltage vs temperature.....	15
Figure 2-27 Normalized current limit vs junction temperature.....	15
Figure 2-28 Step response current limit	15
Figure 4-1 SOT-223 package dimensions	17
Figure 5-1 Recommended furnace temperature curve.....	19
Figure 6-1Recommended pad size	20

1 Block diagram and pin description

1.1 Block diagram

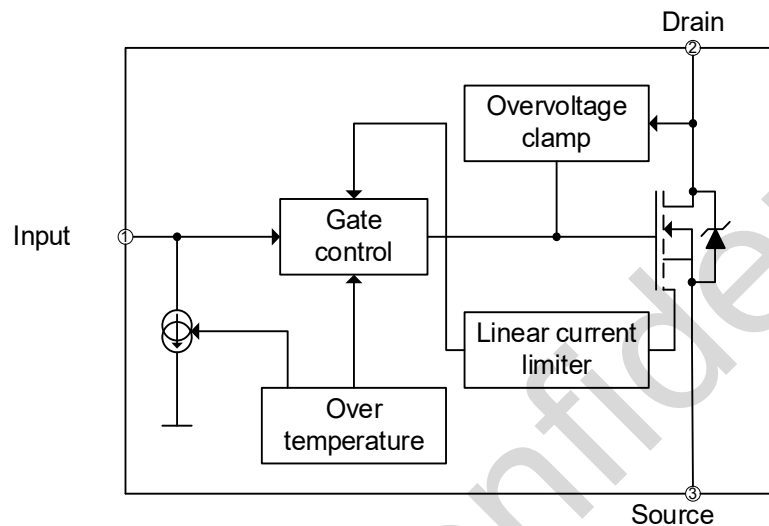


Figure 1-1 Block diagram

SOT-223 Package

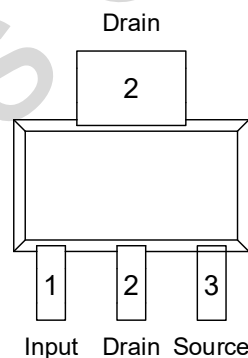


Figure 1-2 Configuration diagram (top view)

2 Electrical specifications

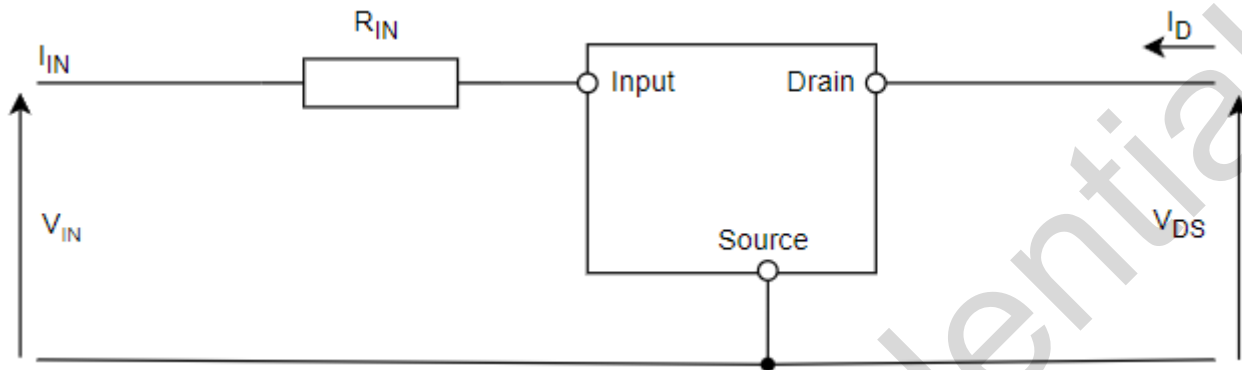


Figure 2-1 Current and voltage conventions

2.1 Absolute maximum ratings

Table 2-1 Absolute maximum ratings

Symbol	Parameter	Value	Unit
V_{DS}	Drain-Source Voltage ($V_{IN} = 0\text{ V}$)	Internally clamped	V
V_{IN}	Input voltage	Internally clamped	V
I_{IN}	Input current	+/- 20	mA
$R_{IN\text{ MIN}}$	Minimum input series impedance	220	Ω
I_D	Drain current	Internally limited	A
I_R	Reverse DC output current	-5.5	A
V_{ESD1}	Electrostatic discharge ($R = 1.5\text{ K}\Omega$, $C = 100\text{ pF}$)	4000	V
P_{tot}	Total dissipation at $T_C = 25^\circ\text{C}$	7	W
T_j	Operating junction temperature	Internally limited	$^\circ\text{C}$
T_c	Case operating temperature	Internally limited	$^\circ\text{C}$
T_{stg}	Storage temperature	-55 to 150	$^\circ\text{C}$

2.2 Thermal data

Table 2-2 Thermal data

Symbol	Parameter	Value	Unit
$R_{thj\text{-case}}$	Thermal resistance junction-case max	18	$^\circ\text{C/W}$
$R_{thj\text{-amb}}$	Thermal resistance junction-ambient max	70 ⁽¹⁾	$^\circ\text{C/W}$

1. When mounted on a standard single-sided FR4 board with 50 mm² of Cu (at least 35 μm thick) connected to all DRAIN pins.

2.3 Electrical characteristics

$-40^{\circ}\text{C} < T_j < 150^{\circ}\text{C}$, unless otherwise specified.

Table 2-3 Off

Symbol	Parameter	Condition	Min	Typ	Max	Unit
V_{CLAMP}	Drain-source clamp voltage	$V_{IN} = 0\text{ V}; I_D = 1.5\text{ A}$	40	44.5	55	V
V_{CLTH}	Drain-source clamp threshold voltage	$V_{IN} = 0\text{ V}; I_D = 2\text{ mA}$	36	44	-	V
V_{INTH}	Input threshold voltage	$V_{DS} = V_{IN}; I_D = 1\text{ mA}$	0.5	1.6	2.5	V
I_{ISS}	Supply current from input pin	$V_{DS} = 0\text{ V}; V_{IN} = 5\text{ V}$	-	87	150	μA
V_{INCL}	Input-source clamp voltage	$I_{IN} = 1\text{ mA}$	6	6.6	7	V
		$I_{IN} = -1\text{ mA}$	-1.0	-	-0.3	V
I_{DSS}	Zero input voltage drain current ($V_{IN} = 0\text{ V}$)	$V_{DS} = 13\text{ V}; V_{IN} = 0\text{ V}; T_j = 25^{\circ}\text{C}$	-	-	10	μA
		$V_{DS} = 25\text{ V}; V_{IN} = 0\text{ V}$	-	-	10	μA

Table 2-4 On

Symbol	Parameter	Condition	Min	Typ	Max	Unit
$R_{DS(on)}$	Static drain-source on resistance	$V_{IN} = 5\text{ V}; I_D = 1.5\text{ A}; T_j = 25^{\circ}\text{C}$	-	-	120	m Ω
		$V_{IN} = 5\text{ V}; I_D = 1.5\text{ A}$	-	-	240	m Ω

Table 2-5 Dynamic ($T_j = 25^{\circ}\text{C}$, unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Unit
$g_{fs}^{(1)}$	Forward transconductance	$V_{DD} = 13\text{ V}; I_D = 1.5\text{ A}$	-	5.0	-	S
C_{OSS}	Output capacitance	$V_{DS} = 13\text{ V}; f = 1\text{ MHz}; V_{IN} = 0\text{ V}$	-	150	-	pF

Table 2-6 Switching ($T_j = 25^{\circ}\text{C}$, unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Unit
$t_{d(on)}$	Turn-on delay time	$V_{DD} = 15\text{ V}; I_D = 1.5\text{ A};$ $V_{gen} = 5\text{ V}; R_{gen} = R_{IN\text{ MIN}} = 220\text{ }\Omega$ (see Figure 2-1)	-	100	300	ns
t_r	Rise time		-	400	900	ns
$t_{d(off)}$	Turn-off delay time		-	500	1350	ns
t_f	Fall time		-	200	750	ns
$t_{d(on)}$	Turn-on delay time	$V_{DD} = 15\text{ V}; I_D = 1.5\text{ A};$	-	0.4	1.35	μs

Electrical specifications

t_r	Rise time	$V_{gen}=5V$; $R_{gen}=2.2\text{ K}\Omega$ (see Figure 2-1)	-	0.65	3	μs
$t_{d(off)}$	Turn-off delay time		-	2.6	8	μs
t_f	Fall time		-	1.4	6	μs
$(di/dt)_{on}$	Turn-on current slope	$V_{DD}=15V$; $I_D=1.5A$; $V_{gen}=5V$; $R_{gen}=R_{IN\text{ MIN}}=220\Omega$	-	3	-	A/ μs
Q_g	Total input charge	$V_{DD}=12V$; $I_D=1.5\text{ A}$; $V_{IN}=5V$; $I_{gen}=2.13mA$ (see Figure 2-4)	-	10	-	nC

Table 2-7 Source drain diode ($T_j = 25^\circ\text{C}$, unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Unit
$V_{SD}^{(1)}$	Forward on voltage	$I_{SD} = 1.5\text{ A}$; $V_{IN} = 0\text{ V}$	-	0.82	-	V
t_{rr}	Reverse recovery time	$I_{SD} = 1.5\text{ A}$; $dI/dt = 12\text{ A}/\mu\text{s}$; $V_{DD} = 30\text{ V}$; $L = 200\mu\text{H}$ (see Figure 2-2)	-	107	-	ns
Q_{rr}	Reverse recovery charge		-	37	-	μC
I_{RRM}	Reverse recovery current		-	0.7	-	A

Table 2-8 Protections ($-40^\circ\text{C} < T_j < 150^\circ\text{C}$, unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Unit
I_{lim}	Current limitation range	$V_{IN} = 5\text{ V}$; $V_{DS} = 13\text{ V}$;	2.3	2.67	4	A
I_{lim_th}	Current limitation trigger threshold	$V_{IN} = 5\text{ V}$; $V_{DS} = 13\text{ V}$; Guarantee by design	2	5	9	A
		$V_{IN} = 5\text{ V}$; $V_{DS} = 13\text{ V}$; $T_j = 25^\circ\text{C}$; Guarantee by design	3.5	6	6.5	A
		$V_{IN} = 5\text{ V}$; $V_{DS} = 13\text{ V}$; $T_j = 125^\circ\text{C}$; Guarantee by design	2	-	4	A
t_{dlim}	Step response current limit	$V_{IN} = 5\text{ V}$; $V_{DS} = 13\text{ V}$		10		μs
T_{jsh}	Overtemperature shutdown		150	175	200	$^\circ\text{C}$
T_{jrs}	Overtemperature reset		135	-	-	$^\circ\text{C}$
I_{gf}	Fault sink current	$V_{IN} = 5\text{ V}$; $V_{DS} = 13\text{ V}$; $T_j = T_{jsh}$	10	15	20	mA
E_{as}	Single pulse avalanche energy	Starting $T_j = 25^\circ\text{C}$; $V_{DD} = 24\text{ V}$; $V_{IN} = 5\text{ V}$; $R_{gen} = R_{IN\text{ MIN}} = 220\Omega$; $L = 24\text{ mH}$ (see Figure 2-3 and Figure 2-5)	100	-	-	mJ

1. Pulsed: Pulse duration = 300 μs , duty cycle 1.5%

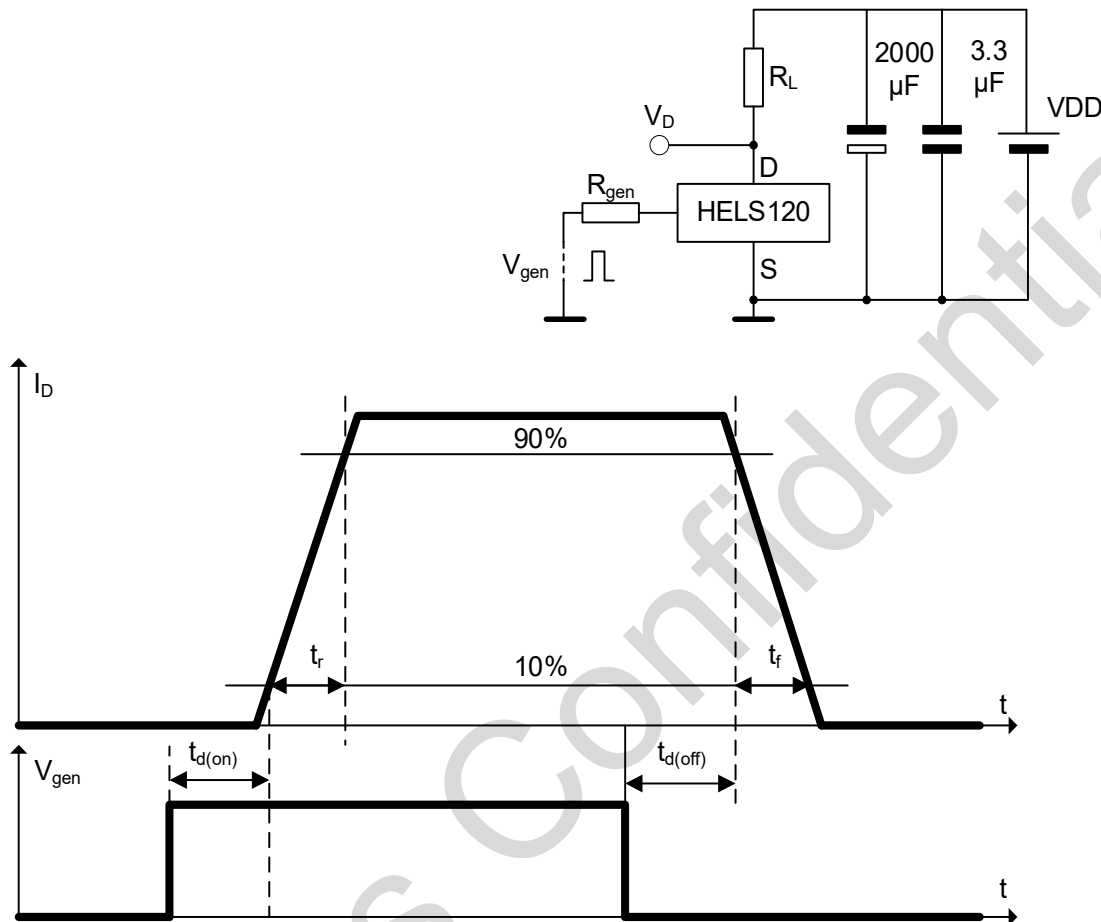


Figure 2-2 Switching time test circuit for resistive load

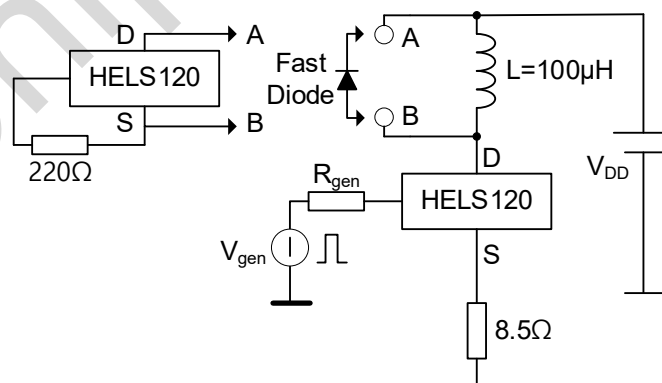


Figure 2-3 Test circuit for diode recovery times

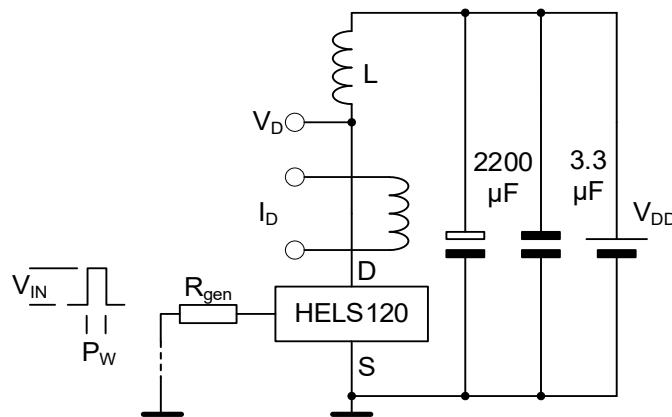


Figure 2-4 Unclamped inductive load test circuits

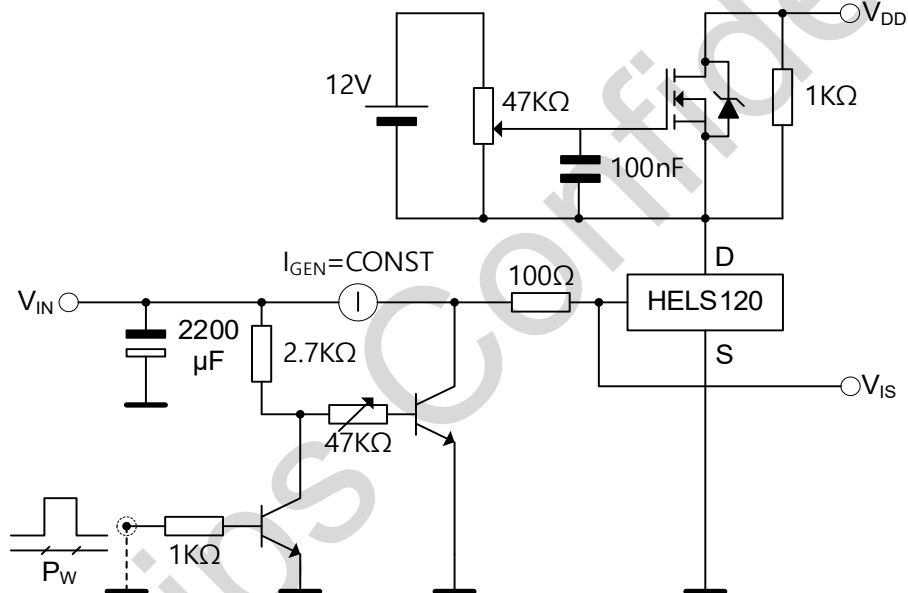


Figure 2-5 Input charge test circuit

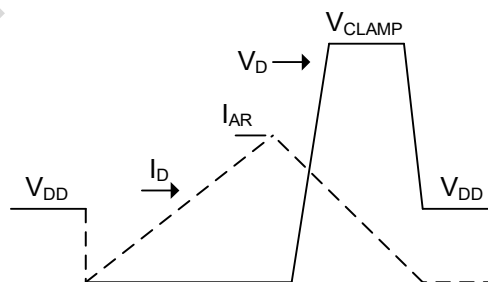


Figure 2-6 Unclamped inductive waveforms

2.4 Electrical characteristics curves

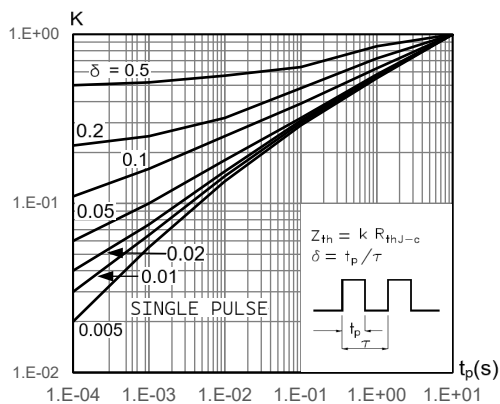


Figure 2-7 Thermal impedance for SOT-223

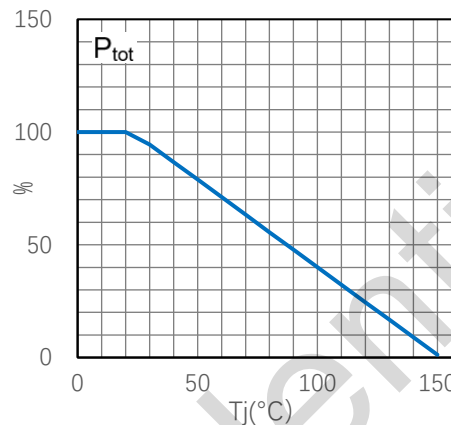


Figure 2-8 Derating curve

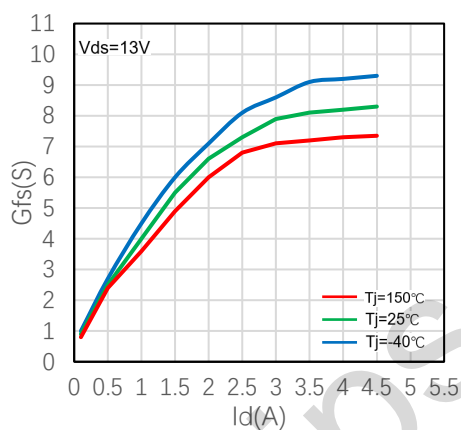


Figure 2-9 Transconductance

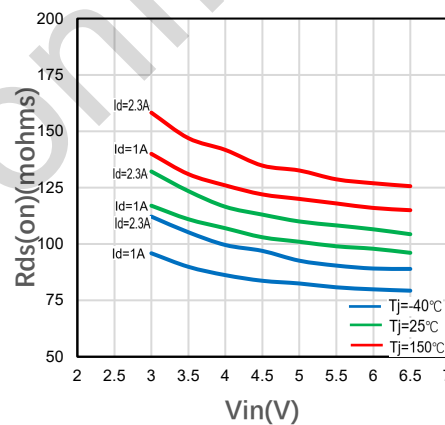


Figure 2-10 Static drain-source on resistance vs input voltage (part 1/2)

Protection feature

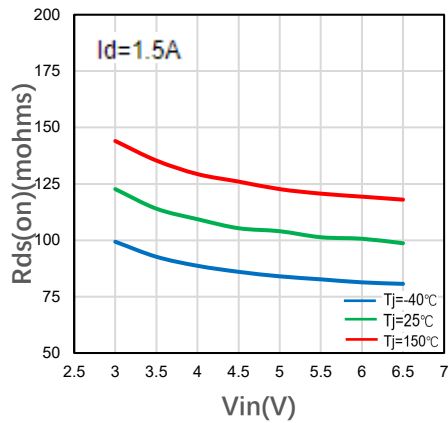


Figure 2-11 Static drain-source on resistance vs input voltage (part 2/2)

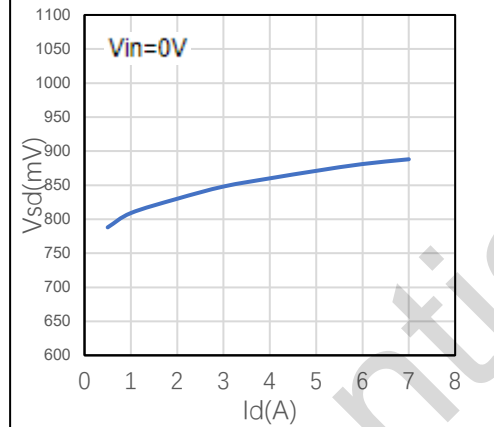


Figure 2-12 Source-drain diode forward characteristics

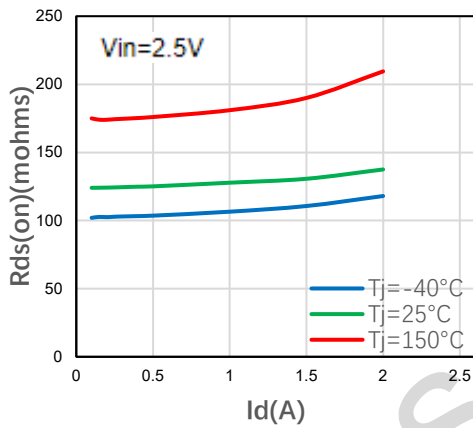


Figure 2-13 Static drain source on resistance

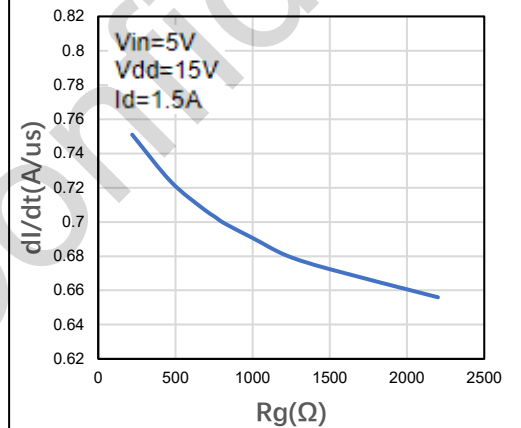


Figure 2-14 Turn-on current slope (part 1/2)

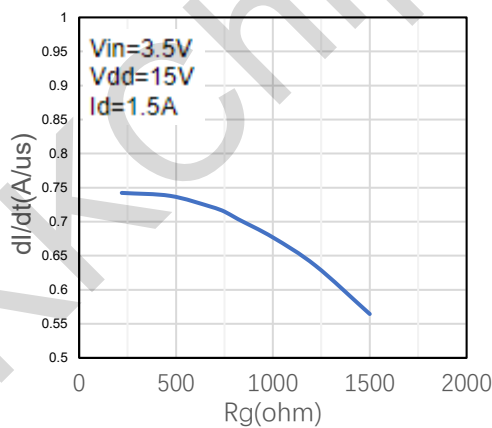


Figure 2-15 Turn-on current slope (part 2/2)

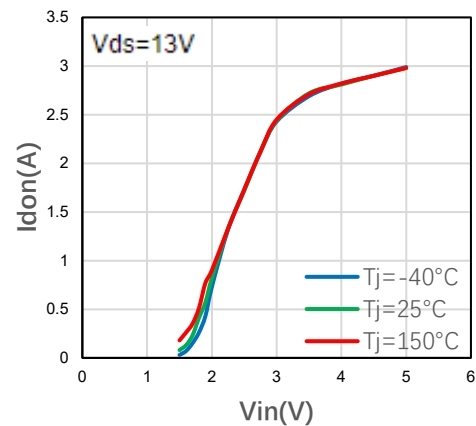


Figure 2-16 Transfer characteristics

Protection feature

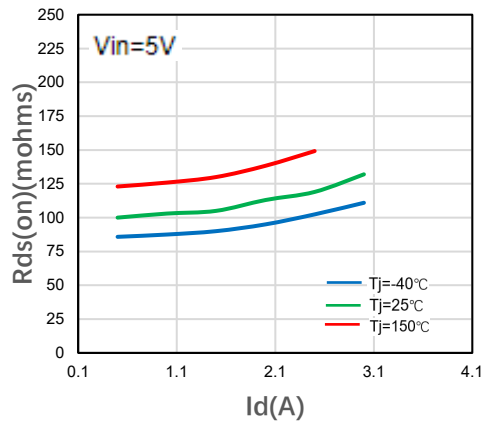


Figure 2-17 Static drain-source on resistance vs Id

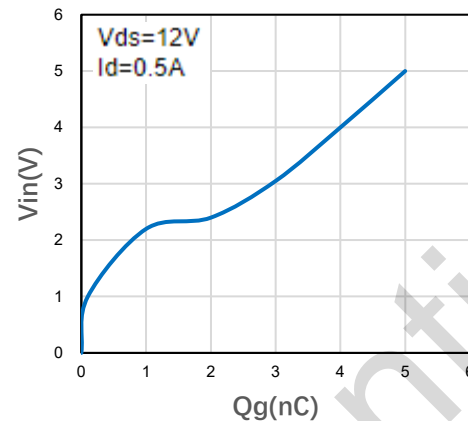


Figure 2-18 Input voltage vs input charge

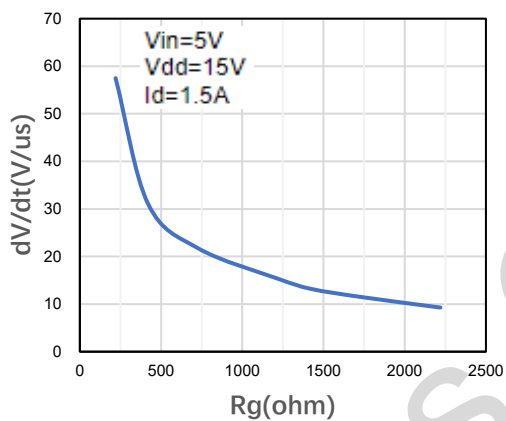


Figure 2-19 Turn-off drain source voltage slope (part 1/2)

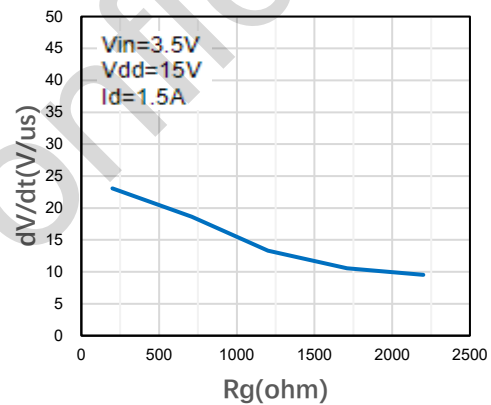


Figure 2-20 Turn-off drain source voltage slope (part 2/2)

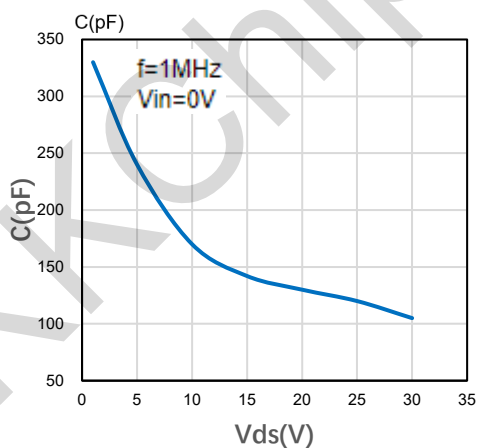


Figure 2-21 Capacitance variations

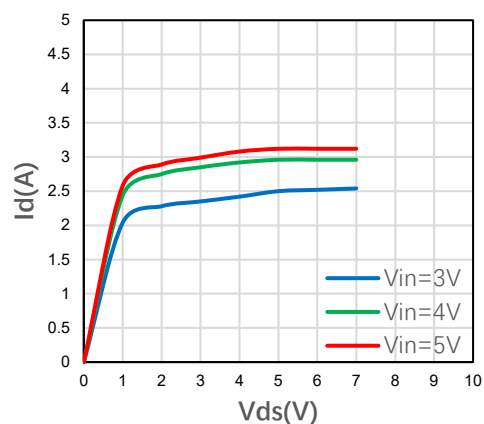


Figure 2-22 Output characteristics temperature

Protection feature

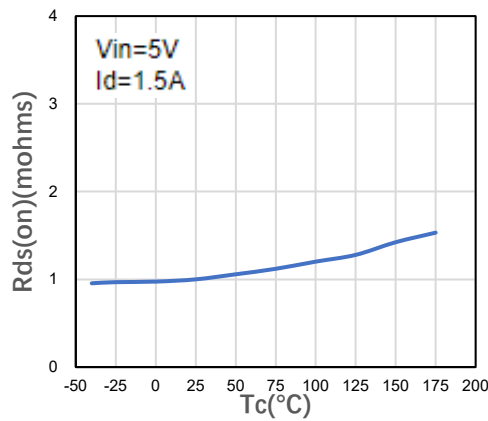


Figure 2-23 Normalized on resistance vs temperature

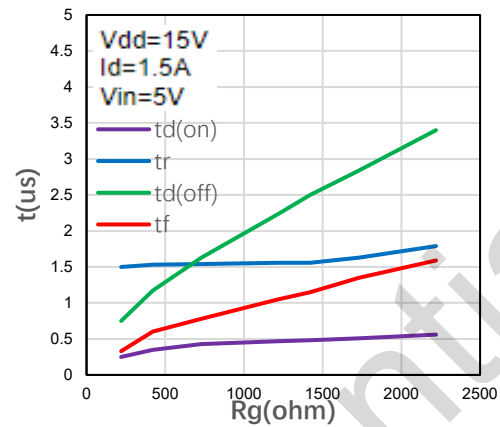


Figure 2-24 Switching time resistive load (part 1/2)

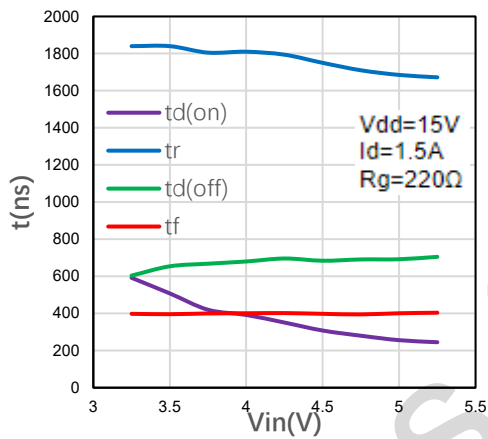


Figure 2-25 Switching time resistive load (part 2/2)

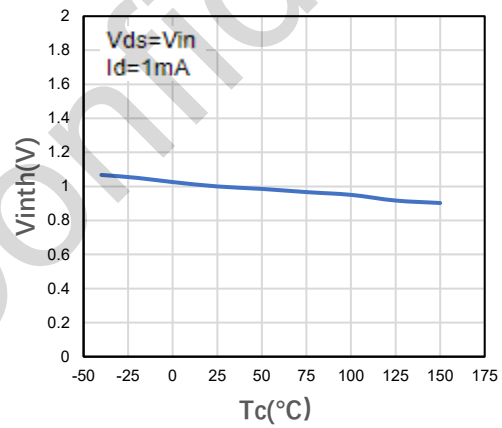


Figure 2-26 Normalized input threshold voltage vs temperature

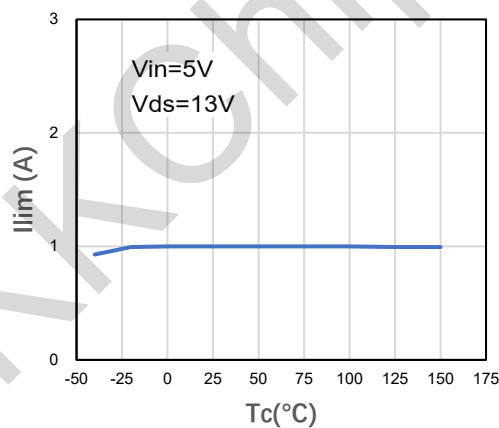


Figure 2-27 Normalized current limit vs junction temperature

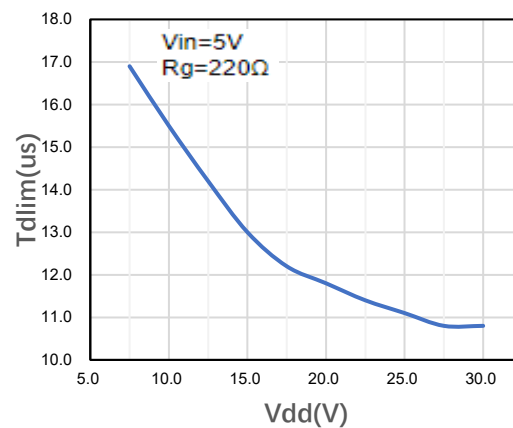


Figure 2-28 Step response current limit

3 Protection features

During normal operation, the input pin is electrically connected to the gate of the internal Power MOSFET through a low impedance path.

The device then behaves like a standard Power MOSFET and can be used as a switch from DC up to 50 kHz. The only difference from the user's standpoint is that a DC current flows into the INPUT pin in order to supply the internal circuitry.

The device integrates:

- Overvoltage clamp protection: internally set at 45 V, along with the rugged avalanche characteristics of the Power MOSFET stage give this device unrivalled ruggedness and energy handling capability. This feature is mainly important when driving inductive loads.
- Linear current limiter circuit: limits the drain current I_D to I_{lim} whatever the input pin voltages. When the current limiter is active, the device operates in the linear region, so power dissipation may exceed the capability of the heatsink. Both case and junction temperatures increase, and if this phase lasts long enough, junction temperature may reach the over temperature threshold T_{jsh} .
- Overtemperature and short circuit protection: these are based on sensing the chip temperature and are not dependent on the input voltage. The location of the sensing element on the chip in the power stage area ensures fast, accurate detection of the junction temperature. Overtemperature cutout occurs in the range 150°C to 190°C, a typical value being 170°C. The device is automatically restarted when the chip temperature falls of about 15°C below shutdown temperature.
- Status feedback: in the case of an overtemperature fault condition ($T_j > T_{jsh}$), the device tries to sink a diagnostic current I_{gf} through the input pin in order to indicate fault condition. If driven from a low impedance source, this current may be used in order to warn the control circuit of a device shutdown. If the drive impedance is high enough so that the input pin driver is not able to supply the current I_{gf} , the input pin will fall to 0 V. This will not however affect the device operation: no requirement is put on the current capability of the input pin driver except to be able to supply the normal operation drive current I_{ISS} .

Additional features of this device are ESD protection according to the Human Body model and the ability to be driven from a TTL logic circuit.

4 Package and packing information

4.1 SOT-223 mechanical data

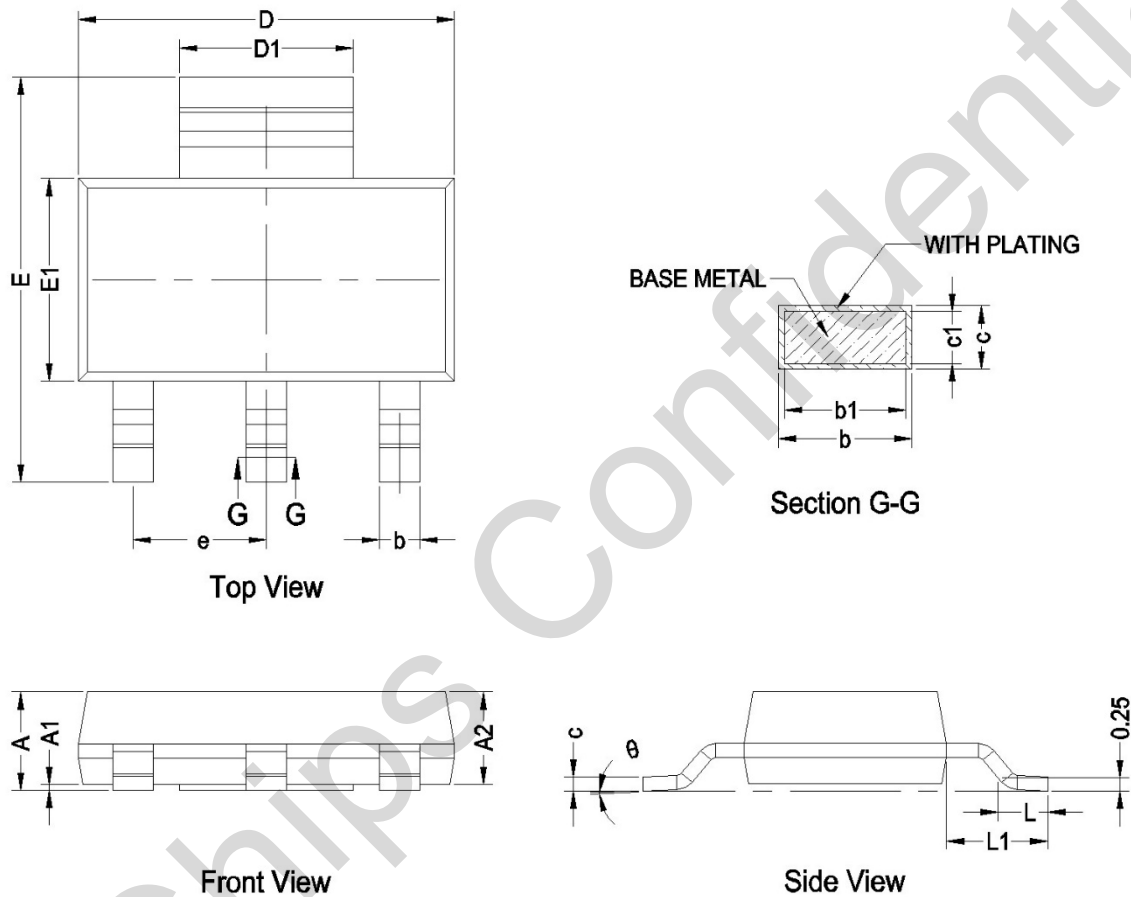


Figure 4-1 SOT-223 package dimensions

Table 4-1 SOT-223 mechanical data

Dim.	mm			Description
	Min.	Typ.	Max.	
A	1.50	1.65	1.80	Total Thickness
A1	0.03	0.06	0.09	Stand Off
A2	1.50	1.60	1.70	Mold Thickness
D1	2.90	3.00	3.10	Lead Width(Plating)
b	0.69	-	0.77	Lead Width(Plating)
b1	0.68	0.71	0.74	Lead Width
c	0.30	-	0.34	L/F Thickness(Plating)
c1	0.30REF			L/F Thickness
D	6.40	6.50	6.60	Body Size(Length)
E	6.80	7.00	7.20	Lead Tip to Tip
E1	3.40	3.50	3.60	Body Size(Width)
e	2.30 BSC			Lead Pitch
L	0.90	-	1.15	Foot Length
θ	0°	-	8°	Foot Angle

5 Recommended furnace temperature curve

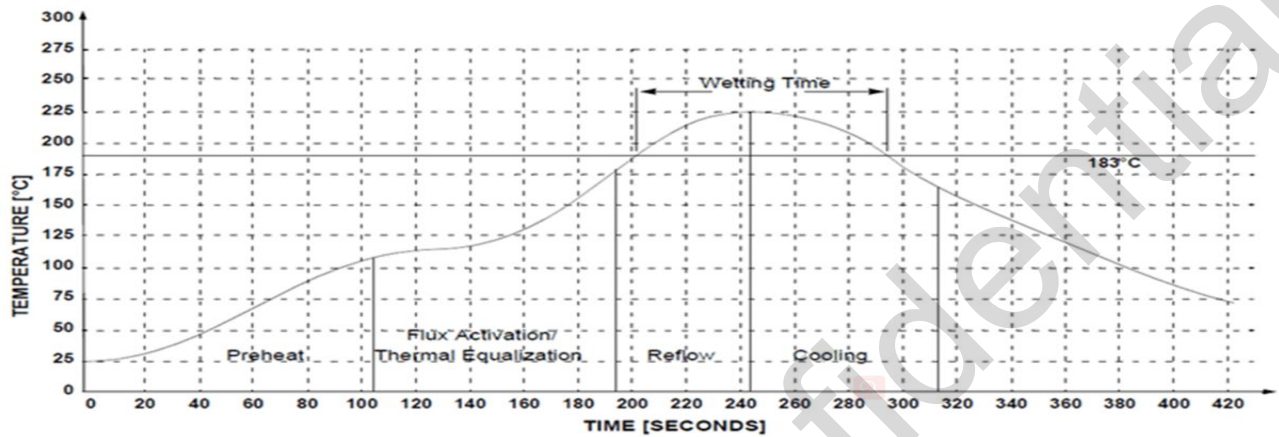


Figure 5-1 Recommended furnace temperature curve

6 Recommended pad size

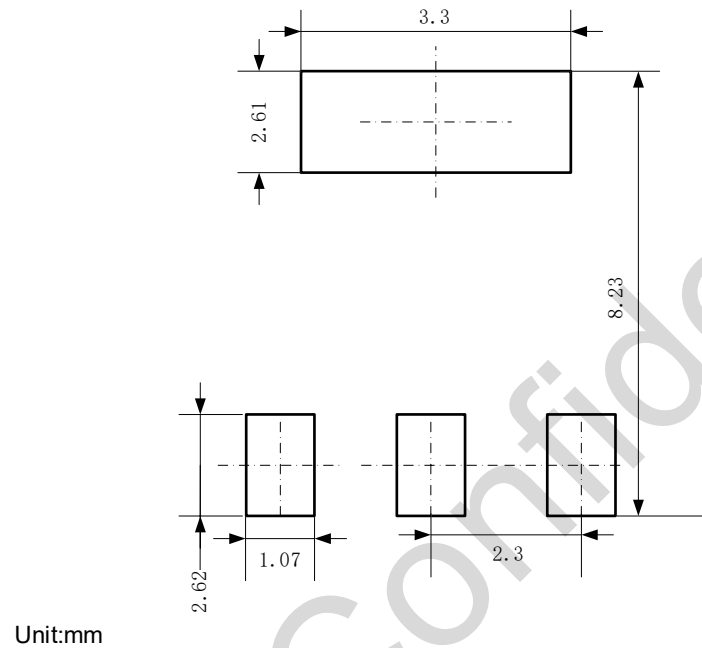


Figure 6-1 Recommended pad size

7 Customer Service

Guangdong KKChips Automotive Electronics Tech Co., Ltd.

9th Floor, Grandtek Building B2, No. 18 Kexue Avenue, Huangpu District, Guangzhou

KKChips Confidential

Revision history

Date	Revision	Changes
2022/10/11	1.1	1. Updated "Note *" in the Section 2 Electrical specifications; 2. Updated Figure 2-1 Current and voltage conventions.
2023/3/13	1.2	1. Update typography; 2. Updated Part number; 3. Update charts; 4. Remove SO-8 related data.
2023/09/12	1.3	1. Format updated to 1.3.
2024/02/05	1.4	1. Format updated to 1.4.
2024/04/24	1.5	1. Package drawing update. 2. Package size update. 3. Format updated to 1.5.
2024/05/11	1.6	1. Change B to D1 in the package diagram.
2024/05/20	1.7	1. Change the package diagram and package information sequence.
2024/07/03	1.8	1. Changed the format of the curve in Section 2.4
2024/09/03	1.9	1. Updated the company's new address
2024/12/18	2.0	1. Format updated to 2.0. 2. Modify tdon_220 parameters from 265ns to 100ns. 3. Modified tr_220 parameters from 1630ns to 400ns, and the maximum value from 1800ns to 900ns. 4. Modified tdoff_220 parameter from 670ns to 500ns, maximum value from 800ns to 1350ns 5. Modified tf_220 parameter from 340ns to 200ns, maximum value from 400ns to 750ns 6. Modified tdon_2200 parameters from 0.6us to 0.4us, maximum value from 0.8us to 1.35us 7. Revised tr_2200 parameters from 1.8us to 0.65us, maximum value from 2us to 3us 8. Modified tdoff_2200 parameters from 3.2us to 2.6us, maximum value from 3.5us to 8us. 9. Revised tf_2200 parameters from 1.6us to 1.4us, maximum value from 2us to 6us 10. Revised dI/dt parameter from 0.7A/us to 3A/us. 11. Revised the description in 3 Protection features.
2025/3/10	2.1	1. Correct error information in Figure 6-1.
2025/4/24	2.2	1. The maximum value of V_{INCL} has been changed from 8V to 7V.
2026/6/3	2.3	1. The Ilimit range is revised from 2.3A~3.3A to 2.3A~4A. 2. Revision Format.